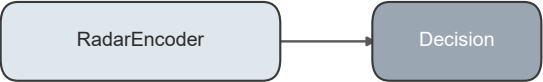


(a) Four architectures compared

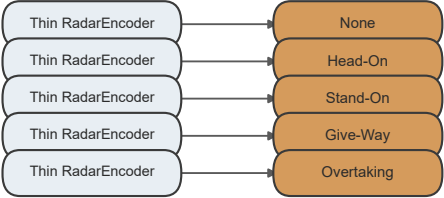
Single network

369K · one perception, one decision



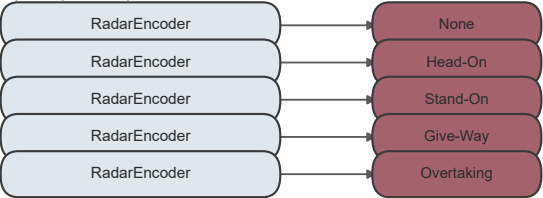
Separate experts, thin

363K · perception duplicated x5, each narrowed



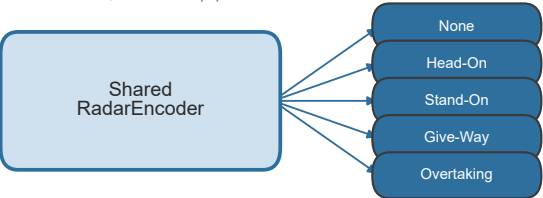
Separate experts, full

1.83M · perception duplicated x5, full width

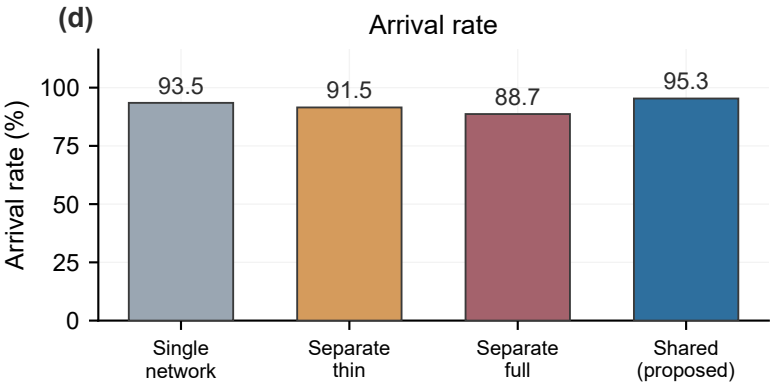
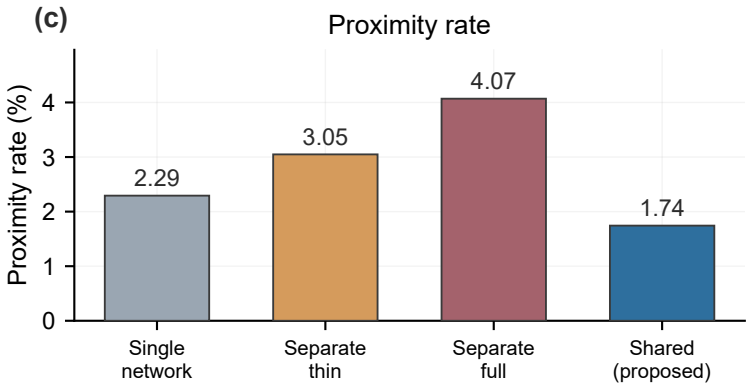
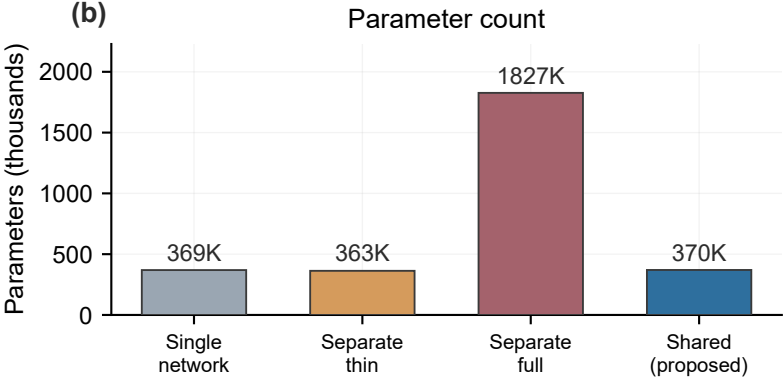


Shared perception (proposed)

370K · shared trunk, residual  $\Delta\mu$  per situation



Radar perception is ~89% of single-network parameters.  
The same expert decomposition is applied to MessageActor, ControlActor and Critic.



Residual shared hub (fc2- $\mu$  shared, per-situation  $\Delta\mu$ ,  $\Delta\mu$  L2) vs single / thin / full, 3 seeds. Proposed hub wins arrival and proximity; parameter count matches the single network. Thin and full still duplicate the radar encoder and lose. THIN/THICK bars are v11 freeze (not residual).